4.1 For an NMOS differential pair with a common-mode voltage V_{CM} applied, as shown in Fig. 4.1, let V_{DD} = 2V, $\mu_n C_{ox}$ = 0.4 mA/V^2, $(W/L)_{1,2}$= 10, V_{TH} = 0.4 V, I_{SS} = 0.16 mA, R_D = 5 kΩ, and neglect channel-length modulation and body effect.

(a) Find V_{OD} and V_{GS} for each transistor.
(b) For V_{CM} = 1V, find V_P , I_{D1}, I_{D2}, V_{OUT1}, and V_{OUT2}.
(c) Repeat (b) for V_{CM} = 1.4 V.
(d) Repeat (b) for V_{CM} = 0.9 V.
(e) What is the highest value of V_{CM} for which M_1 and M_2 remain in saturation?
(f) If current source I_{SS} requires a minimum voltage of 0.2 V to operate properly, what is the lowest value allowed for V_P and hence for V_{CM}?
(g) Find the small-signal differential gain under the condition of (b).

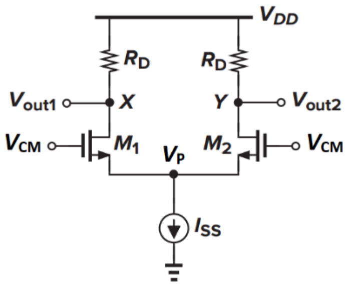


Fig. 4.1

(a)

$$\frac{I_{ss}}{2} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_{1,2} V_{od}^2 \Rightarrow V_{od} = 0.2\ V$$
$$V_{GS} = V_{od} + V_{th} = 0.6\ V \tag{1-1}$$

(b)假设处于饱和区

$$V_P = V_{CM} - V_{GS} = 1 - 0.6 = 0.4\ V$$
$$I_{D1} = I_{D2} = \frac{I_{SS}}{2} = 0.08\ mA$$
$$V_{OUT1} = V_{OUT2} = V_{DD} - I_{D1,2}R_D = 2 - 0.4 = 1.6\ V \tag{1-2}$$
$$V_{DS} = V_{OUT} - V_P = 1.6 - 0.4 = 1.2\ V > V_{od}$$

假设正确

(c) 假设处于饱和区

$$V_P = V_{CM} - V_{GS} = 1.4 - 0.6 = 0.8\ V$$
$$I_{D1} = I_{D2} = \frac{I_{SS}}{2} = 0.08\ mA$$
$$V_{OUT1} = V_{OUT2} = V_{DD} - I_{D1,2}R_D = 2 - 0.4 = 1.6\ V \tag{1-3}$$
$$V_{DS} = V_{OUT} - V_P = 1.6 - 0.8 = 0.8\ V > V_{od}$$

假设正确

(d) 假设处于饱和区

$$V_P = V_{CM} - V_{GS} = 0.9 - 0.6 = 0.3\ V$$
$$I_{D1} = I_{D2} = \frac{I_{SS}}{2} = 0.08\ mA$$
$$V_{OUT1} = V_{OUT2} = V_{DD} - I_{D1,2}R_D = 2 - 0.4 = 1.6\ V \tag{1-4}$$
$$V_{DS} = V_{OUT} - V_P = 1.6 - 0.3 = 1.3\ V > V_{od}$$

假设正确

(e)

$$V_{CM} \leqslant V_{OUT1,2}+V_{TH}=2\ V \tag{1-5}$$

(f)

$$\begin{aligned}&V_P \geqslant 0.2\ V\\&V_{CM} \geqslant V_{p\min}+V_{GS}=0.2+0.6=0.8\ V\end{aligned} \tag{1-6}$$

(g)

$$\begin{aligned}&A_V=-g_m R_D=-4\\&g_m=\sqrt{2I_{1,2}\mu_n C_{ox}\left(\frac{W}{L}\right)_{1,2}}=\frac{1}{1250}\ S\end{aligned} \tag{1-7}$$

4.2 Consider the differential amplifier specified in Problem 4.1 with gate voltage of M2, V_{G2}, fixed to V_{CM} =1V, and V_{G1} is set to $V_{CM}+v_{id}$. Let v_{id} be adjusted to the value that causes I_{D1} =0.09 mA and I_{D2} =0.07 mA.

(a) Find the corresponding values of V_{GS2}, V_P, V_{GS1}, and hence v_{id}.

(b) What is the difference output voltage $v_{od}=V_{out2}-V_{out1}$?

(c) What is the voltage gain v_{od}/v_{id}? Explain the reason why the gain obtained here is different with that obtained in 4.1 (g).

(d) What value of v_{id} results in I_{D1} = 0.07 mA and I_{D2} = 0.09 mA?

(e) What is the maximum range of v_{id} that ensures either I_{D1} or I_{D2} be larger than 0.

(a)

$$\begin{aligned}&V_{GS1}=\sqrt{\frac{2I_{D1}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_1}}+V_{TH}=0.212+0.4=0.612\ V\\&V_{GS2}=\sqrt{\frac{2I_{D2}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_2}}+V_{TH}=0.187+0.4=0.587\ V\\&V_P=V_{G1}-V_{GS1}=1-0.612=0.388\ V\\&v_{id}=V_{GS1}-V_{GS2}=0.612-0.587=0.025\ V\end{aligned} \tag{2-1}$$

(b)假设处于饱和区

$$\begin{aligned}&V_{OUT1}=V_{DD}-I_{D1}R_D=2-0.45=1.55\ V\\&V_{OUT2}=V_{DD}-I_{D2}R_D=2-0.35=1.65\ V\\&v_{od}=V_{OUT1}-V_{OUT1}=0.1\ V\end{aligned} \tag{2-2}$$

处于饱和区假设成立

(c)

$$\frac{v_{od}}{v_{id}}=\frac{0.1}{0.025}=4 \tag{2-3}$$

(d)

$$v_{id}=-0.025\ V \tag{2-4}$$

(e)

$$v_{id} \leqslant \sqrt{\frac{2I_{SS}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{1,2}}}=0.28\ V \tag{2-5}$$

4.3 Fig. 4.3 shows a MOS differential amplifier with the drain resistors R_D implemented using diode-connected PMOS transistors, M3 and M4. Let M1 and M2 be matched, and M3 and M4 be matched. Neglect body effect.

(a) Draw the differential half-circuit and use it to derive an expression for A_{DM} in terms of $g_{m1,2}$, $g_{m3,4}$, $r_{o1,2}$, and $r_{o3,4}$.

(b) Neglecting the effect of the output resistances r_o, find A_{DM} in terms of $\mu_n C_{ox}$, $\mu_p C_{ox}$, $(W/L)_{1,2}$ and $(W/L)_{3,4}$.

(c) With the condition of (b), if $\mu_n=4\mu_p$ and all four transistors have the same channel length, find $(W_{1,2}/W_{3,4})$ that results in A_{DM} = 10 V/V.

(d) Find the maximum $V_{in,CM}$ that ensures saturation of M1 and M2 in terms of V_{DD}, I_{SS}, $\mu_p C_{ox}$, $(W/L)_{3,4}$, V_{THP} and V_{THN}.

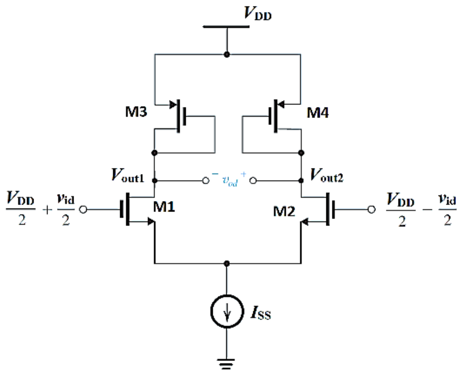


Fig. 4.3

(a)

等效电路略

$$A_{DM}=g_{m1,2}\left(\frac{1}{g_{m3,4}}\parallel r_{o3,4}\parallel r_{o1,2}\right) \tag{3-1}$$

(b)

$$A_{DM}=\frac{g_{m1,2}}{g_{m3,4}}=\sqrt{\frac{\mu_n C_{ox}\left(\frac{W}{L}\right)_{1,2}}{\mu_p C_{ox}\left(\frac{W}{L}\right)_{3,4}}} \tag{3-2}$$

(c)

$$10=2\sqrt{\frac{W_{1,2}}{W_{3,4}}}\Longrightarrow\frac{W_{1,2}}{W_{3,4}}=25 \tag{3-3}$$

(d)

$$V_{in,CM}\leqslant V_{OUT}+V_{TH1,2}=V_{DD}-V_{od3,4}+V_{TH1,2}=V_{DD}-\sqrt{\frac{I_{SS}}{\mu_p C_{ox}\left(\frac{W}{L}\right)_{3,4}}}+V_{THN} \tag{3-4}$$

(e)

4.4 For the differential amplifier shown in Fig. 4.4, neglect the channel length modulation effect.

(a) Draw the differential half-circuit for the amplifier and use it to derive an expression for the differential gain $A_{DM} \equiv v_{od} / v_{id}$ in terms of g_m, R_D, and R_S.

(b) What is the gain with R_S =0?

(c) What is the value of R_S (in terms of $1/g_m$) that reduces the gain to half of value at R_S=0?

(d) Derive the maximum range of v_{id} (in terms of g_m, I_{SS}, R_S) that ensures either I_{D1} or I_{D2} be larger than 0.

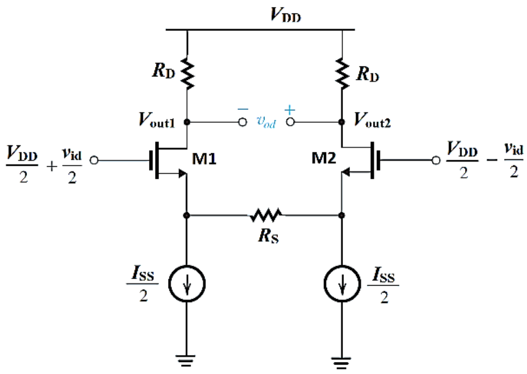
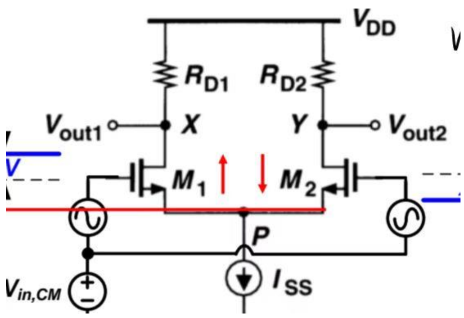


Fig. 4.4

(a) 见ppt39页

$$A_v = \frac{g_{m1} R_D}{1 + g_{m1} R_s / 2} \tag{4-1}$$

(b) 短路为基础的差动放大器



$$A_v = g_{m1} R_D \tag{4-2}$$

(c)

$$R_s = 2 / g_m \tag{4-3}$$

(d)V_{in1}刚好截止时，

$$V_{in1} - V_P = V_{TH} \tag{4-4}$$

此时V_{in2}流过两路电流，

$$V_{in2} - V_P = R_S \times \frac{I_{SS}}{2} + V_{TH} + \sqrt{\frac{2 I_{SS}}{\mu_n C_{ox} \frac{W}{L}}} \tag{4-5}$$

相减可得

$$|V_{in1} - V_{in2}| = \frac{R_S I_{SS}}{2} + \sqrt{\frac{2 I_{SS}}{\mu_n C_{ox} \frac{W}{L}}} = \frac{R_S I_{SS}}{2} + \sqrt{2} \cdot V_{OD} \tag{4-6}$$

4.5 A design error has resulted in a gross mismatch in the circuit of Fig. 4.5. Specifically, M2 has twice the W/L ratio of M1. Neglect channel length modulation effect and body effect. If v_{id} is a small AC signal, find:

(a) I_{D1} and I_{D2} for v_{id}=0.

(b) The offset voltage, $v_{od,offset}$, at the output nodes. (Hint: $v_{od,offset}=V_{out1}-V_{out2}$ when v_{id}=0.)

(c) V_{OD} for each of M1 and M2 in terms of I_{SS}, W/L and $\mu_n C_{ox}$.

(d) The differential gain A_{DM} in terms of R_D, I_{SS}, and V_{OD}. (Note: you cannot use the half-circuit method as the circuit is asymmetrical)

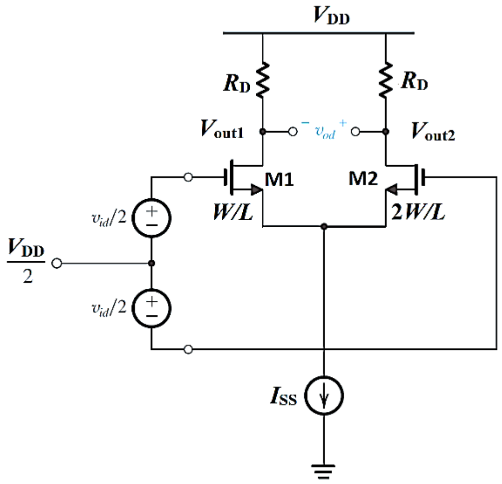


Fig. 4.5

(a)

$$\begin{cases} I_2 = 2I_1 \\ I_1 + I_2 = I_{SS} \end{cases} \Rightarrow \begin{cases} I_1 = \frac{1}{3} I_{SS} \\ I_2 = \frac{2}{3} I_{SS} \end{cases} \tag{5-1}$$

(b)

$$v_{od,offset} = (I_2 - I_1) R_D = \frac{1}{3} I_{SS} \tag{5-2}$$

(c)

$$\begin{aligned} V_{od1} &= \sqrt{\frac{2I_1}{\mu_n C_{ox}\left(\frac{W}{L}\right)_1}} = \sqrt{\frac{2I}{3\mu_n C_{ox}\left(\frac{W}{L}\right)_1}} \\ V_{od2} &= \sqrt{\frac{2I_2}{\mu_n C_{ox}\left(\frac{W}{L}\right)_2}} = \sqrt{\frac{4I}{3\mu_n C_{ox}\left(\frac{W}{L}\right)_2}} \end{aligned} \tag{5-3}$$

(d)

$$\begin{aligned} A_{DM} &= \frac{2R_D}{\frac{1}{g_{m1}} + \frac{1}{g_{m2}}} \\ g_{m1} &= \frac{2I_1}{V_{OD1}} = \frac{2I_{SS}}{3V_{OD1}} \\ g_{m2} &= \frac{2I_2}{V_{OD2}} = \frac{4I_{SS}}{3V_{OD2}} \end{aligned} \tag{5-4}$$

4.6 For the differential amplifier shown in Fig. 4.6, let M1 and M2 have $\mu_P C_{OX}$ = 0.4 mA/V^2, W/L=10, and assume that the bias current source has an output resistance of 30 kΩ. Neglect channel length modulation effect and body effect

(a) Find V_{OD}, g_m, the differential-mode gain A_{DM} and the common-mode gain A_{CM}.

(b) If the drain resistances are known to have a mismatch of 2%, e.g. R_{D1}=4kΩ*(1+1%) and R_{D1} = 4kΩ*(1-1%), find the $A_{CM\text{-}DM}$, and the CMRR in dB (i.e. CMRR_dB=20log$_{10}$(A_{DM}/$A_{CM\text{-}DM}$)).

(c) On the basis of the condition given in (b), if the width of the M1 and M2 have a mismatch of 2% further, find $A_{CM\text{-}DM}$, and CMRR (in dB) again.

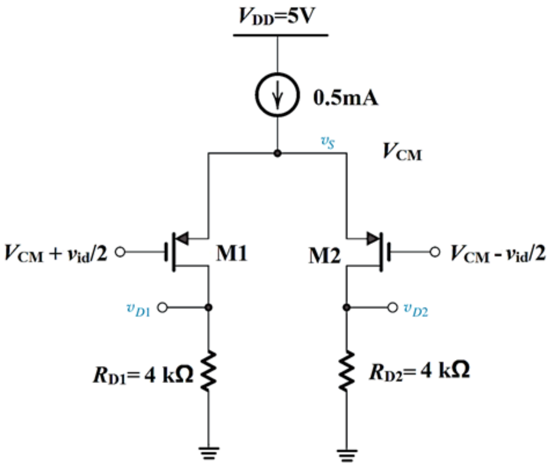


Fig. 4.6

(a)

$$V_{od1,2} = \sqrt{\frac{2I_{1,2}}{\mu_p C_{ox}\left(\frac{W}{L}\right)_{1,2}}} = \sqrt{\frac{0.5}{0.4\times 10}} = 0.353553\ V$$

$$g_{m1,2} = \frac{2I_{1,2}}{V_{od}} = \frac{0.5}{0.353553} = 1.41422\ mS \tag{6-1}$$

$$A_{DM} = g_m R_D = 5.66$$

$$A_{CM} = \frac{\frac{R_D}{2}}{\frac{1}{2g_m} + R_{SS}} = 0.06589$$

(b)

$$A_{CM-DM} = \frac{g_m \Delta R_D}{1 + 2g_m R_{SS}} = 1.3178\times 10^{-3}$$

$$CMRR = \frac{g_m R_D}{\frac{g_m \Delta R_D}{1+2g_m R_{SS}}} = \frac{R_D}{\Delta R_D}(1+2g_m R_{SS}) = 4292.66\ = 72.65\ \ dB \tag{6-2}$$

(c)

$$A_{CM-DM} = \frac{\Delta g_m R_D}{1 + 2g_m R_{SS}} = 1.3178\times 10^{-3}$$

$$CMRR = \frac{g_m R_D}{\frac{\Delta g_m R_D}{1+2g_m R_{SS}}} = \frac{g_m}{\Delta g_m}(1+2g_m R_{SS}) = 4292.66\ = 72.65\ \ dB \tag{6-3}$$